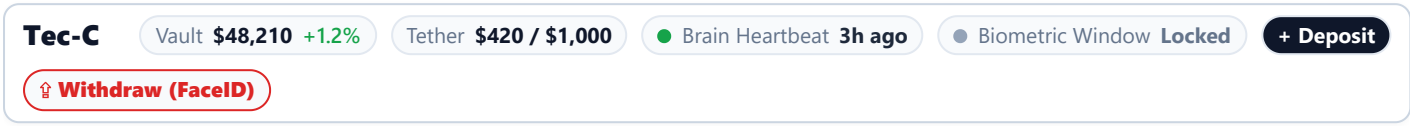
TSINGHUA PBCSF & WEBANK • FINTECHATHON 2026 • TOPIC F: AGENT WALLET

Tec-C: Complete System, UI Architecture & Operational Guide

The Autonomous Corporate Expense Card for Web3 AI Agents with Cryptographic Safety Invariants

The Big Picture: Traditional trading bots require your master private keys, meaning one prompt injection or smart contract bug can drain 100% of your funds. **Tec-C (Tethered Expense Corporate Card)** gives the AI a policy-tethered corporate card: it trades autonomously 24/7 inside a user-configured daily leash, but is mathematically blocked on-chain from exceeding limits, touching unvetted tokens, or withdrawing your treasury.



01 The 3 Pillars of Tec-C

1. Non-Custodial Agent Wallet

The Smart Safe. Built on ERC-7579 & ERC-4337. Assets stay in the smart vault. The AI never sees or holds your master key—it holds an ephemeral session key restricted by bytecode hooks.

2. Mathematical Leash

The Hard Spending Boundary. Enforced on-chain by `TetherGuardHook.sol`. Daily limits are converted to 8-decimal USD via Chainlink price feeds. The AI physically cannot exceed the limit.

3. Mobile Virtual Session

The Master Remote Key. Runs on your laptop, but connects securely to your smartphone via Cloudflare tunnel. Monitor positions and approve breakout trades via FaceID from any coffee shop.

02 The Core Problem Solved: Autonomy vs. Safety

Feature / Risk	Traditional Trading Bots	Standard Hardware Wallet	Tec-C Autonomous Vault
Autonomous 24/7 Trading	✓ Full autonomy	✕ Zero (Manual click for everything)	✓ Full autonomy inside your limit
Catastrophic Hack Risk	✕ High (API keys can drain 100%)	✓ Low (Keys kept offline)	✓ Zero (Capped by strict on-chain leash)
Daily Limit Tuning	✕ Unbounded	✕ Not applicable	✓ User configures initial limit & opt-in elasticity
Multi-Chain Vault Operations	✕ Fragmented / complex	✕ Manual network switching	✓ Multi-chain deposit & Biometric withdrawal
Privacy Before Cloud AI	✕ Leaks full balances & addresses	✓ N/A	✓ Privacy Sieve (Normalized 1,000 privacy units)

03 Dynamic Elasticity: The Accuracy Feedback Loop

Personalized Limits: AI Earns or Loses Autonomy Based on Accuracy — Onboarding Consent

During onboarding, you choose your starting daily allowance (e.g. $1,000/day). You are then asked: "*Do you want your limit to be personalized and adaptively influenced by AI accuracy?*"

✓ High Accuracy (Trust Rewards): When the AI makes profitable trades and you ratify them in the Morning Review, the vault awards a step-up (up to **+10% per day max**) towards your configured ceiling.

✕ Mistakes (Immediate Penalties): If the AI enters a bad pool, flagging it in Morning Review contracts the daily limit by **-20%**. Adding a penalty note contracts it by **-10%**. Trust is lost 7x faster than it is gained!

04 Screen 1: Home & Command Center

The central hub where you submit structured investment mandates, monitor the live pipeline, and toggle the continuous background arbitrage engine.

1. New Command Composer — Interactive

5 structured building blocks translate your intent into a typed mathematical mandate (no prompt injection vectors):

1 · Action: `Find opportunities` | `Invest` | `Stake` | `Swap` | `Rebalance` | `Reduce risk` | `Exit position` | `Park idle cash`

2 · Market: `Crypto (Ethereum)` | `Chinese market (FISCO BCOS)` | `Commodities (Arbitrum One)` | `Any (All enabled)`

3 · Asset / Scope: `ETH` | `stETH` | `rETH` | `USDC` | `All in market`

4 · Capital Amount: `5%` | `10%` | `25%` | `Custom Slider` (e.g. *10% = $4,821 · uses $482 of remaining Tether*)

5 · Risk Tier: `Tier 1 (Core)` | `Tier 2 (Growth)` | `Tier 3 (Unrestricted - Locked)`

"Stake ETH and stETH in Crypto, using 10% of capital, Tier 2." ➔ ***[Submit to Brain]***

2. Live Pipeline Tracker — Request #0412

- ● **Command:** Submitted (0.2 s)
- ● **Brain Analysis:** 3 options discovered (1.4 s)
- ◐ **Ratings:**
  - News & Sentiment: Done (+0.71)
  - Econometrics & Liquidity: Running
  - Injection Defense: Waiting
- ○ **Brain Decision:** Awaiting synthesis
- ○ **Executor:** Session key dispatch
- ○ **On-Chain Guard:** Tether check
- ○ **Morning Review:** Pending

Dedicated Background Arbitrage Engine (PegGuard) — Independent Toggle Switch

Arbitrage is NOT a one-off trade command inside the composer. It is a standalone, persistent toggle switch located outside the command box:

- **When ENABLED [ON]:** The bot operates 24/7 in the background via `POST /api/arbitrage/toggle`, continuously scanning the 24-asset catalog across 4 categories (Class B Fiat Stablecoins, Class A LSTs/LRTs, Class A Wrapped BTC, Class B RWAs) spanning Ethereum L1, Arbitrum One L2, and FISCO BCOS. Features *Stateful Unprofitable Watermark Memory* to prevent wasting gas on stagnant prices.
- **When DISABLED [OFF]:** The arbitrage scanner is completely discarded and idle, conserving gas and network activity until re-enabled.

05 Multi-Chain Vault Operations & Biometric Security

Multi-Chain Deposit — All Supported Networks

Deposit funds seamlessly on any available connected chain:
- **Ethereum Mainnet & Arbitrum One** (ETH, USDC, USDT)
- **FISCO BCOS Consortium Chain** (e-CNY, enterprise tokens)
- **Optimism & Polygon** (DeFi collateral)

Deposits credit your normalized 8-decimal USD vault balance via Chainlink price feeds.

Biometric Withdrawal — FaceID / TouchID Required

Strict Security Invariant: The AI session key has ZERO authority to withdraw funds to external addresses.
- Withdrawing capital to an external address **strictly requires hardware biometric verification (FaceID / TouchID via WebAuthn NIST P-256)**.
- Even if the AI agent server is fully breached, the user's treasury can never be drained.

06 Screen 2: Opportunities & Bounded AI Rating

When a command is submitted, the AI Brain explores multiple paths and presents a transparent, graded comparison with full risk breakdowns.

Zone A · Candidates for Request #0412 (Stake ETH / stETH, 10% Capital)

Option A · Stake ETH on Lido — Chosen 70%

Crypto · Staking · 2.0 ETH · $4,821

News: **+0.71** | Econ: **0.64** | Injection: **Pass**

0.68 Weighted Score

Option B · Supply ETH on Aave — Chosen 30%

Crypto · Lending · 2.0 ETH · $4,821

News: **+0.55** | Econ: **0.70** | Injection: **Pass**

0.61 Weighted Score

Option C · Buy stETH on DEX — Rejected

Crypto · Swap · 2.0 ETH · $4,821

News: **+0.71** | Econ: **0.41** | Injection: **Warn**

0.44 Weighted Score

Zone B · Deep Rating Breakdown for Option A (Lido Staking)

1. News & NLP Sentiment — +0.71 Positive

- **FinBERT Sentiment:** Scores from -1 (Panic) to +1 (Euphoria). Current: **+0.71**.
- **Recent Headlines:** 4 articles processed in last 9h (3 Positive, 1 Neutral).
- **Vault Sovereignty Invariant:** News informs Brain trade sizing only; it can never freeze or halt the smart vault.

2. Econometrics & Liquidity — Acceptable

- **Tail Risk (CVaR):** 95% CVaR is -3.1% | 99% CVaR is -5.4%.
- **Slippage Model:** Modeled curve estimates 0.08% slippage for a $4,821 execution.
- **Liquidity Comparison:**

Staking: 3.4% APY	CVaR: -3.1%	Slip: 0.08%
Lending: 2.6% APY	CVaR: -2.4%	Slip: 0.02%

3. Prompt-Injection Defense — Pass

- ✓ **Whitelist:** Target contract verified.
- ✓ **Honeypot Test:** Clean token bytecode.
- ✓ **Contract Age:** 4.2 years battle-tested.
- ✓ **Amount Check:** Matches command mandate.
- ✓ **Instruction Scan:** Zero injection payloads.

Zone C · Brain Decision & Autonomous Synthesis

Agent Weights (Adjustable in Settings)

News 30% | Econometrics 45% | Injection 25%

Synthesis Reason: Options A and B both pass hard filters with close scores (0.68 vs 0.61). The Brain chooses a portfolio diversification split.

Execution Outcome — Autonomous

Split: A 70% ($3,375) · B 30% ($1,446)

Inside today's Tether Radius ($482 used of $580 left), so it executes autonomously without requiring manual button clicks.

Other Possible Outcomes: Proceed with Single Best • Do Nothing (Negative Edge) • Needs Biometric Breakout (Approve / Reject)

07 Screen 3: Portfolio Holdings & Hourly Watchdog

Comprehensive multi-asset treasury monitoring combined with an automated hourly safety scanner.

Total Portfolio Valuation **+1.2% in 24h**

$48,210 +$574 Today

Multi-chain balances normalized to 8-decimal USD via Chainlink price feeds.

Asset Allocation **4 Sectors**

- **Crypto:** 52%
- **Chinese Mkt:** 22%
- **Commodities:** 16%
- **Arbitrage:** 10%

Live Treasury Holdings

Asset	Market / Chain	Balance	USD Value	Share	24h P&L	News Sentiment	Health Status
stETH	Crypto (Ethereum)	6.20	$14,933	31%	**+4.1%**	Positive (+0.71)	● **OK**
USDC	Crypto (Arbitrum One)	10,200	$10,200	21%	0.0%	Neutral (0.00)	● **OK**
PAXG	Commodities (Arbitrum)	3.10	$7,720	16%	**+1.8%**	Neutral (+0.12)	● **Watch**
e-CNY MMF	Chinese Market (FISCO BCOS)	10,600	$10,600	22%	**-0.6%**	**Negative (-0.45)**	● **Re-evaluate**

08 The Hourly Watchdog Engine

🛡 24-Hour Autonomous Health Scanner **Next scan in 38m**

The Watchdog runs automatically every 60 minutes via `/trigger_hourly_monitor` across all held positions. It inspects on-chain pool depth, oracle feeds, and financial news signals:

●●●●●●●●●●●●●●● Clean Scans (22) ● Market Warning (1) ● Negative News Alert (1)

Active Watchdog Alert: *e-CNY MMF · negative news detected* ➔ Brain triggered a risk-reduction proposal to de-risk 10% of position into USDC collateral.

09 Screen 4: Morning Review & Adaptive Learning

Every morning, you spend 15 seconds reviewing what the AI traded while you were asleep. This is where you directly reward, correct, and train your AI.

Overnight Trades: **5** Capital Moved: **$1,240** Reviewed: **2 of 5** Ratify the Rest

📈 **Like Streak Toward Radius Increase** **3 of 5 Approved · Eligible for +10% Step-Up**

Trade #1: Stake 0.4 ETH on Lido

Executed at 02:14 UTC · Crypto Market · $964 Value

Pending Review

Slippage: **0.08%** (est. 0.10%) P&L Now: **+$3.20** Tx Hash: `0x9a4f...e1c8`

👍 Like & Ratify | 👎 Flag for Revert (-20%) | 📝 Note

Your Note to the Brain (Opens on " 📝 Note" click): 12 / 25 words

"Lido staking is approved, but prefer rocket pool (rETH) when yield difference is under 0.2%."

⚠ Penalize Autonomy (-10%) Sent to AI Brain memory → Send Note

How AI Brain Acts: Notes are transmitted to the Brain's learning engine (`tecc_brain.learning.notes`). The AI updates its internal protocol affinity, refines cost multipliers, and initiates proactive portfolio adjustments or rebalancing if required.

Swap 500 USDC to PAXG · 03:40 UTC · Commodities Pending Review ˅

Supply 1 ETH to Aave v3 · 01:02 UTC · Crypto ✓ Ratified · Allowance Restored

10 Real-World Scenarios & Everyday Life

How all systems operate together during regular days, high-value opportunities, and market crises:

Scenario A: Silent Daily Yield Harvesting

100% Autonomous

Situation: You set an initial daily allowance of $1,000 for your vault.
AI Discovery: The AI Brain spots Compound v3 offering an 11.2% lending yield on USDC.
Execution: The AI deposits $800 into Compound. Because $800 is under your $1,000 leash, it executes silently on-chain. You wake up with extra yield without having to click a button.

Scenario B: The High-Value Breakout Trade

Biometric FaceID Required

Situation: The AI Brain spots a flash $4,500 stablecoin arbitrage capturing a $120 net profit in 30 seconds.
What Happens: The smart contract detects that $4,500 exceeds your $1,000 daily leash. It blocks autonomous execution and buzzes your smartphone.
User Action: You glance at the notification card, verify the profit, scan FaceID. The 5-minute window opens and the trade completes instantly.

Scenario C: Dynamic Elasticity in Action

Adaptive Learning

Starting Point: You configure a $1,000 daily limit with Dynamic Elasticity enabled (Hard Ceiling: $2,500).
Week 1 (High Accuracy): The AI executes 5 successful yield trades. You hit "Ratify" in Morning Review. The vault rewards it by +10%/day, expanding allowance to $1,460.
Week 2 (Correction): The AI enters an asset you dislike. You flag it. The vault penalizes it by -20%, shrinking the limit down to $1,168.

Scenario D: PegGuard Crisis Circuit-Breaker

Depeg Defense

Situation: You hold 6,000 units of a synthetic coin. Breaking news indicates backing reserve distress; price slips to $0.982.
Automated Defense: Within seconds, PegGuard activates its emergency circuit-breaker and swaps the 6,000 coins into native USDC. An hour later the coin collapses to $0.40. Your capital was saved.

Scenario E: Remote Coffee Shop Access (Virtual Session)

Encrypted Tunnel

Setup: You click "Create Virtual Session" on your home PC and scan the QR code with your iPhone.
Experience: An encrypted tunnel connects to your phone browser. Authenticated via phone FaceID, you steer risk, monitor trades, and approve breakouts all day without opening ports.

11 Summary of Unbreakable Security Invariants

1. AI can NEVER withdraw funds: External transfers strictly require hardware FaceID.

2. AI can NEVER exceed daily limit: Leash enforced on-chain via Chainlink USD feeds.

3. AI can NEVER touch unwhitelisted tokens: Bytecode whitelist blocks fake tokens.

4. Trust is lost 7x faster than gained: -20% penalty on flag vs. max +10% daily growth cap.